An Asymptotic Equivalence of Choice of Model by Cross-validation and Akaike's Criterion

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SUMMARY

A logarithmic assessment of the performance of a predicting density is found to lead to asymptotic equivalence of choice of model by cross-validation and Akaike's criterion, when maximum likelihood estimation is used within each model.

Keywords: PREDICTING DENSITY; MODEL CHOICE; AKAIKE'S INFORMATION CRITERION; CROSS-VALIDATION

1. INTRODUCTION

AKAIKE (1973) proposed a criterion for model choice equivalent to the following: If α indexes the model, choose α to maximize

$$L(\alpha, \hat{\theta}_\alpha) - p_\alpha, \tag{1.1}$$

where $L(\alpha, \theta_\alpha)$ is the log-likelihood function, $\hat{\theta}_\alpha$ is the maximum likelihood estimate of the parameter θ_α in the model α and p_α is the dimensionality of θ_α.

Akaike's derivation of (1.1) was for hierarchical models but, as he finally remarked, this restriction is unnecessary. Looking at (1.1), we see p_α as a correction term without which we would be maximizing $L(\alpha, \hat{\theta}_\alpha)$; models with parameters of high dimensionality are given a severe handicap by this correction term.

For normal multiple linear regression models with known variance, σ^2, Mallows' C_p (Gorman and Toman, 1966) is given by

$$C_p = (\mathrm{RSS}_\alpha/\sigma^2) - (n - 2p_\alpha), \tag{1.2}$$

where RSS_α is the residual sum of squares for model α and n is the sample size. From (1.2) we see that maximizing (1.1) is equivalent to minimizing C_p.

Akaike's criterion stemmed from a recognition that unreserved maximization of likelihood provides an unsatisfactory method of choice between models that differ appreciably in their parametric dimensionality. Since the method of cross-validatory choice (Stone, 1974) is also concerned with the latter problem, it is perhaps unsurprising that a relationship can be established between the two approaches.

2. THE CHOICE PROBLEM

Adopting the notation of Stone (1974), we suppose we have a data-base

$$S = \{(x_i, y_i), i = 1, \ldots, n\}$$

for n items and that our problem is the choice of predicting density for y given x from a prescribed class of formal predicting densities

$$\{f(y \mid x, \alpha, S), \alpha \in \mathscr{A}\}, \tag{2.1}$$

whose members are indexed by the choice parameter α. All densities for y are with respect to a common fixed measure with generic element dy. The operational interpretation of (2.1) is that the choice of α specifies a predicting density of y for each x, whose form depends in a prescribed way on S. The notation is not intended to carry any other probabilistic interpretation.

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It is useful to distinguish two complementary cases of (2.1):

Case 1. $f(y|x, \alpha, S) = f(y|x, \alpha)$ independent of S;
Case 2. $f(y|x, \alpha, S)$ properly dependent on S.

In Case 1, (2.1) becomes formally equivalent to a statistical model with α as conventional parameter. In Case 2, our attention will be focused on a general example which we will call Example A after Akaike (1973). Its prescription is

$$f(y|x, \alpha, S) \equiv f_\alpha(y|x, \hat{\theta}_\alpha(S)), \tag{2.2}$$

where

$$\{f_\alpha(y|x, \theta_\alpha),\ \theta_\alpha \in \Theta_\alpha\} \tag{2.3}$$

are the densities for a conventional parametric model α and $\hat{\theta}_\alpha(S)$ is the supposed unique maximum likelihood estimator maximizing $L(\alpha, \theta_\alpha) = \sum_i \log f_\alpha(y_i|x_i, \theta_\alpha)$.

3. Log-density Assessment

Suppose $f^{(i)}(y)$, $i = 1, \ldots, n$, were presented as predicting densities for y_i, $i = 1, \ldots, n$, respectively. As a measure of their success, take the log-density assessment

$$A = \sum_i \log f^{(i)}(y_i). \tag{3.1}$$

Observe that A is the logarithm of $\prod_i f^{(i)}(y_i)$ which may be termed the predicting probability density evaluated at the observations.

For Case 1, use of $f^{(i)}(y) = f(y|x_i, \alpha)$, $i = 1, \ldots, n$, would have the assessment

$$A(\alpha) = \sum_i \log f(y_i|x_i, \alpha), \tag{3.2}$$

whence we see that choice of α to maximize $A(\alpha)$ would be equivalent to maximum likelihood "estimation" of α for the "log-likelihood" given by the right-hand side of (3.2). Thus Case 1 introduces no innovations.

For Case 2, it would be unrealistic to assess the choice of α with $f^{(i)}(y) = f(y|x_i, \alpha, S)$ because S itself contains y_i. It is more realistic to use the cross-validatory

$$f^{(i)}(y) = f(y|x_i, \alpha, S_{-i})$$

where $S_{-i} = S - (x_i, y_i)$. This gives us

$$A(\alpha) = \sum_i \log f(y_i|x_i, \alpha, S_{-i}). \tag{3.3}$$

We will show in the next section that for Example A, $A(\alpha)$, given by (3.3), is asymptotically equivalent, under weak conditions, to Akaike's criterion (1.1), which, as we have seen, "corrects" maximum likelihood as a method of choice of model.

4. Asymptotic Equivalence

For simplicity, we treat α as fixed and omit it from the notation. Writing l for $\log f$, with f given by (2.2) and (2.3), A in (3.3) equals $\sum_i l(y_i|x_i, \hat{\theta}(S_{-i}))$. With $L(\theta) = \sum_j l(y_j|x_j, \theta)$, we have that $\hat{\theta}(S)$ [$\hat{\theta}$ for short] maximizes $L(\theta)$ and $\hat{\theta}(S_{-i})$ [$\hat{\theta}_{-i}$ for short] maximizes $L(\theta) - l(y_i|x_i, \theta)$. We *suppose* that $\theta = (\theta_1 \ldots \theta_p)^{\mathrm{T}} \in \Theta$ an open region of R^p and that f is twice-differentiable with respect to θ. Write

$$l' = \left(\frac{\partial l}{\partial \theta_1} \cdots \frac{\partial l}{\partial \theta_p}\right)^{\mathrm{T}}, \quad l'' = \left(\frac{\partial^2 l}{\partial \theta_i \partial \theta_j}\right)$$

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with similar notation for L. We *suppose* that $\hat{\theta}$ and $\hat{\theta}_{-i}$ are unique solutions of $L'(\theta) = \mathbf{0}$ and $L'(\theta) - l'(y_i|x_i, \theta) = \mathbf{0}$ respectively. Then by Taylor's theorem

$$A = L(\hat{\theta}) + \sum_i (\hat{\theta}_{-i} - \hat{\theta})^{\mathrm{T}} l'\{y_i|x_i, \hat{\theta} + a_i(\hat{\theta}_{-i} - \hat{\theta})\}, \tag{4.1}$$

$$L'(\hat{\theta}_{-i}) = L''\{\hat{\theta} + b_i(\hat{\theta}_{-i} - \hat{\theta})\}(\hat{\theta}_{-i} - \hat{\theta}) \tag{4.2}$$

with $|a_i| \leqslant 1$, $|b_i| \leqslant 1$, $i = 1, \ldots, n$. Also

$$L'(\hat{\theta}_{-i}) = l'(y_i|x_i, \hat{\theta}_{-i}). \tag{4.3}$$

From (4.1), (4.2) and (4.3), *supposing* L'' in (4.2) is invertible,

$$A = L(\hat{\theta}) + \sum_i l'(y_i|x_i, \hat{\theta}_{-i})^{\mathrm{T}} [L''\{\hat{\theta} + b_i(\hat{\theta}_{-i} - \hat{\theta})\}]^{-1} l'\{y_i|x_i, \hat{\theta} + a_i(\hat{\theta}_{-i} - \hat{\theta})\}. \tag{4.4}$$

Next *suppose* that S is a random sample from some joint distribution P of (x, y). Let E denote expectation with respect to P. With this supposition we can expect:

(i) $\hat{\theta} \xrightarrow{\mathrm{P}} \theta_0$ as $n \to \infty$ where θ_0 is the *supposed* unique value of θ maximizing $E\{l(y|x, \theta)\}$;

(ii) $\hat{\theta}_{-i} \xrightarrow{\mathrm{P}} \theta_0$ as $n \to \infty$ for $i = 1, 2, \ldots$;

(iii) $n^{-1} L''(\hat{\theta} + b_i(\hat{\theta}_{-i} - \hat{\theta})) \xrightarrow{\mathrm{P}} E\{l''(y|x, \theta_0)\} = L_2$, say;

(iv) $n^{-1} \sum_i l'\{y_i|x_i, \hat{\theta} + a_i(\hat{\theta}_{-i} - \hat{\theta})\} l'(y_i|x_i, \hat{\theta}_{-i})^{\mathrm{T}} \xrightarrow{\mathrm{P}} E\{l'(y|x, \theta_0) l'(y|x, \theta_0)^{\mathrm{T}}\} = L_1$, say.

So we have, heuristically, established that A is asymptotically

$$L(\hat{\theta}) + \mathrm{trace}(L_2^{-1} L_1). \tag{4.5}$$

Since θ_0 maximizes $E\{l(y|x, \theta)\}$, it follows that $E\{l''(y|x, \theta_0)\}$ is negative-definite. Hence the correction term in (4.5), written in the form $E\{l'(y|x, \theta_0)^{\mathrm{T}} L_2^{-1} l'(y|x, \theta_0)\}$ is seen to be negative. However, little more can be said about it without further assumptions of a statistical character. The key assumption that gives us our asymptotic equivalence with Akaike's criterion is: *The conditional distribution of y given x in the distribution P is $f(y|x, \theta^*)$ for some unique $\theta^* \in \Theta$, that is, the conventional model $\{f(y|x, \theta), \theta \in \Theta\}$ is true.* In fact, this assumption implies $\theta^* = \theta_0$. For

$$E\{l(y|x, \theta_0)\} = E\left\{\int f(y|x, \theta^*) \log f(y|x, \theta_0)\, dy\right\}$$

$$\leqslant E\left\{\int f(y|x, \theta^*) \log f(y|x, \theta^*)\, dy\right\} = E\{l(y|x, \theta^*)\}$$

and θ_0 is the supposed unique maximizer of $E\{l(y|x, \theta)\}$. Further, differentiating the identity $\int f(y|x, \theta) l'(y|x, \theta)\, dy = \mathbf{0}$ with respect to θ, setting $\theta = \theta_0$ and taking expectations with respect to x, we find $L_1 = -L_2$ (the well-known identity). Hence the correction term in (4.5) is $\mathrm{trace}(-I_{p \times p}) = -p$ and asymptotically

$$A = L(\hat{\theta}) - p \tag{4.6}$$

which is identical to (1.1) once the missing α's are restored.

While the key assumption italicized above gives us the general equivalence, weaker assumptions will suffice for particular choices of $\{f_\alpha(y|x, \theta_\alpha), \theta_\alpha \in \Theta_\alpha\}$.

If we consider two models α_1, α_2 of type (2.3) with

$$\Theta_{\alpha_1} \subset \Theta_{\alpha_2}$$

and suppose that both are true, then it is well known that, under regularity conditions, $2\{L(\alpha_2, \hat{\theta}_{\alpha_2}) - L(\alpha_1, \hat{\theta}_{\alpha_1})\}$ is asymptotically χ^2 with $d = p_{\alpha_2} - p_{\alpha_1}$ degrees of freedom. Hence, by

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(4.6), $A(\alpha_2)-A(\alpha_1)$ is asymptotically $\frac{1}{2}\chi^2_d-d$. This shows how the simpler model will be favoured by the choice criterion $A(\alpha)$.

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REFERENCES

AKAIKE, H. (1973). Information theory and an extension of the maximum likelihood principle. *2nd International Symposium on Information Theory* (B. N. Petrov and F. Czáki, eds), pp. 267–281. Budapest: Akademiai Kiadó.

GORMAN, J. W. and TOMAN, R. J. (1966). Selection of variables for fitting equations to data. *Technometrics*, **8**, 27–51.

STONE, M. (1974). Cross-validatory choice and assessment of statistical predictions (with Discussion). *J. R. Statist. Soc.* B, **36**, 111–147.

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